ML07 Tables and reading a CSV file

Week 3, Tables and statistics | Session 7 of 24 | Machine Learning | New library: pandas

Today's question. How many sales are in the practice shop file, and what does each one record?

Objective. By the end: you can read a CSV file into a pandas table and say how many rows and columns it has.

> **WHERE THIS IS USED**
> A shop till, a clinic register and a school timetable are all tables with rows and columns. Programs read such tables from files before they can learn anything. Which table in your life would you like to open in Python?

Before this session: ML06 Functions and checks that fail loudly. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
row and column	A row is one record across the table. A column is one kind of fact down the table.	One line of a register and one heading of the register.
header	The name written at the top of each column.	The title above a column in an exercise book.
CSV file	A plain text file where each line is a row and commas split the columns.	A register copied as text.
library and import	A library is code other people wrote. import loads it into your notebook.	Borrowing a toolbox from the workshop.
DataFrame	The pandas name for a table inside Python.	A spreadsheet that lives in the notebook.
shape	The number of rows and columns, written as a pair.	Counting how many lines and headings a register has.

Block A. Build a small table

pandas is a library for tables. Run the cell and look at the table it prints.

Predict. Predict how many rows and columns the table has. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
sales = pd.DataFrame({"item": ["Book", "Pen", "Book"], "units": [2, 5, 3]})
print(sales)
print(sales.shape)
```

Run it, then compare. The table has 3 rows and 2 columns, so shape prints (3, 2). The headers are item and units. import pandas as pd loads the library and gives it the short name pd. A table made this way is a DataFrame. The dot after sales reaches a property that belongs to that table.

Investigate.

1. What does the first number of shape count?
2. Where do the headers come from?

Modify. Add a fourth sale, Pen with 4 units, to both lists and run again.

Block B. Read the practice shop file

camp_shop.csv sits beside your notebook. Read it into a table and look at the first rows.

Predict. Predict how many rows camp_shop.csv has and which headers it shows. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
print(shop.head())
print(shop.shape)
print(list(shop.columns))
```

Run it, then compare. The table has 60 rows and 5 columns, so shape prints (60, 5). The headers are sale_id, date, item, units and unit_price_ugx. read_csv reads the file into a DataFrame and head shows its first five rows. The unit of money is in the header: unit_price_ugx is in shillings.

Investigate.

1. Why is the unit written in the header name unit_price_ugx?
2. What do the 60 rows stand for?

Modify. Show the first 10 rows by writing 10 inside the brackets of head.

Make

Read synthetic_homes.csv. Show its first rows and its shape. Say what each row stands for and what unit rent_ugx is in.

Save your work as ML07. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
FileNotFoundError: No such file or directory	The CSV file is not beside the notebook.	Move the file into the same folder as the notebook, or ask the instructor.
ModuleNotFoundError: No module named 'pandas'	The prepared camp environment is not active.	Tell the instructor. Do not install packages.

Exit ticket

A CSV file becomes a table of rows and columns, and shape tells me how big it is.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.